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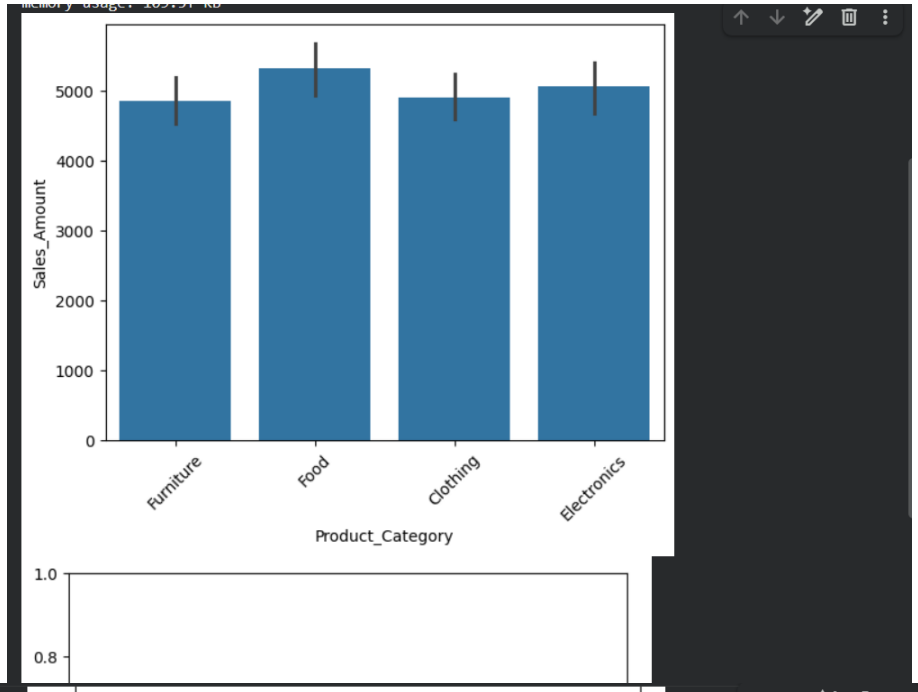
Task 1 : **Data Cleaning & Visualization Project**

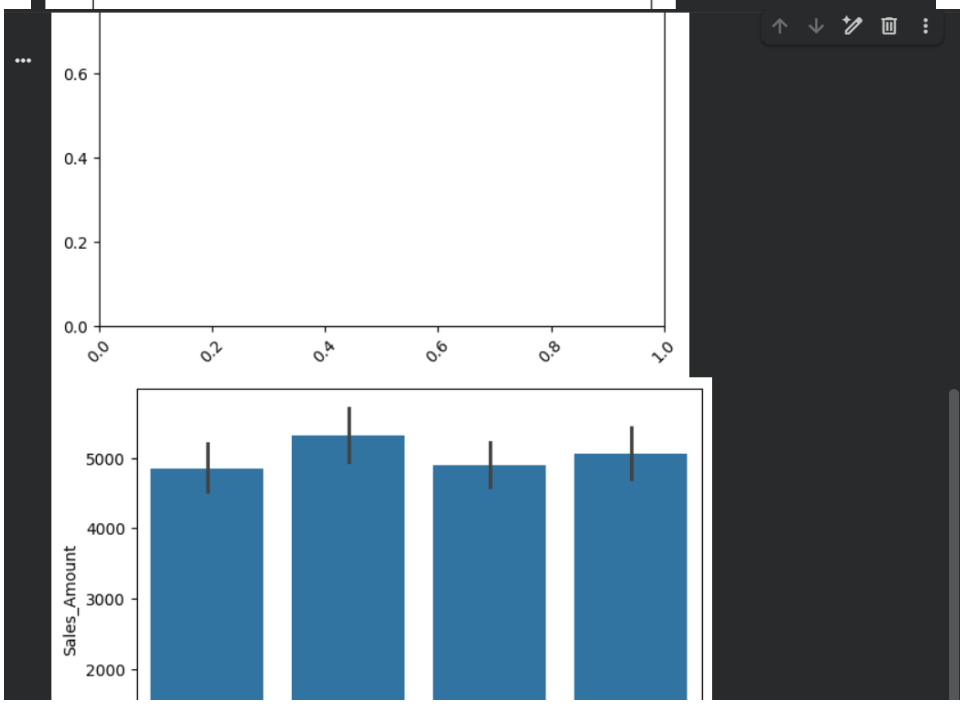
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload()

df = pd.read_csv("sales_data.csv")
df.head()

df = df.dropna()

df = df.drop_duplicates()

df.info()
df.describe()

sns.barplot(x='Product_Category', y='Sales_Amount', data=df)
plt.xticks(rotation=45)
plt.show()
plt.xticks(rotation=45)
plt.show()

sns.barplot(x='Product_Category', y='Sales_Amount', data=df)
plt.xticks(rotation=45)
plt.show()

sns.barplot(x='Region', y='Sales_Amount', data=df)
plt.xticks(rotation=45)
plt.show()

df['Sale_Date'] = pd.to_datetime(df['Sale_Date'])

sns.lineplot(x='Sale_Date', y='Sales_Amount', data=df)
plt.xticks(rotation=45)
plt.show()

df.columns

sns.heatmap(df.corr(), annot=True)
plt.show()
```

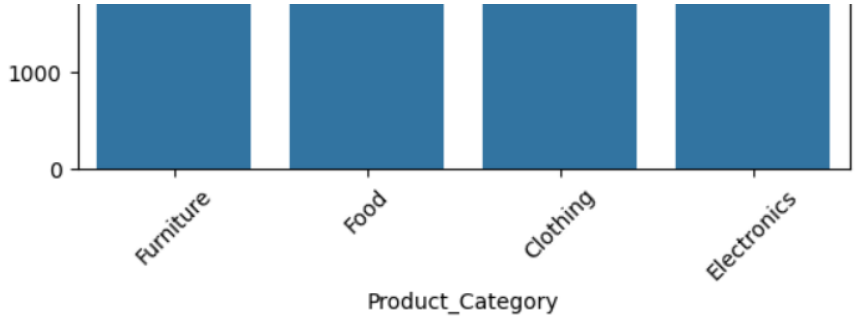
Choose Files sales_data.csv

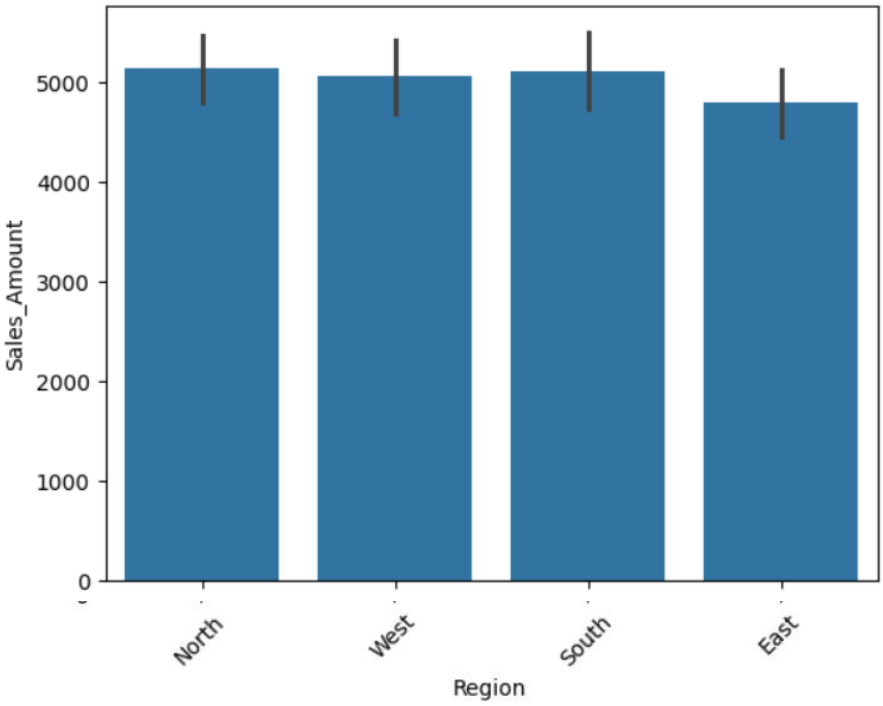
sales_data.csv(text/csv) - 105083 bytes, last modified: 4/20/2026 - 100% done

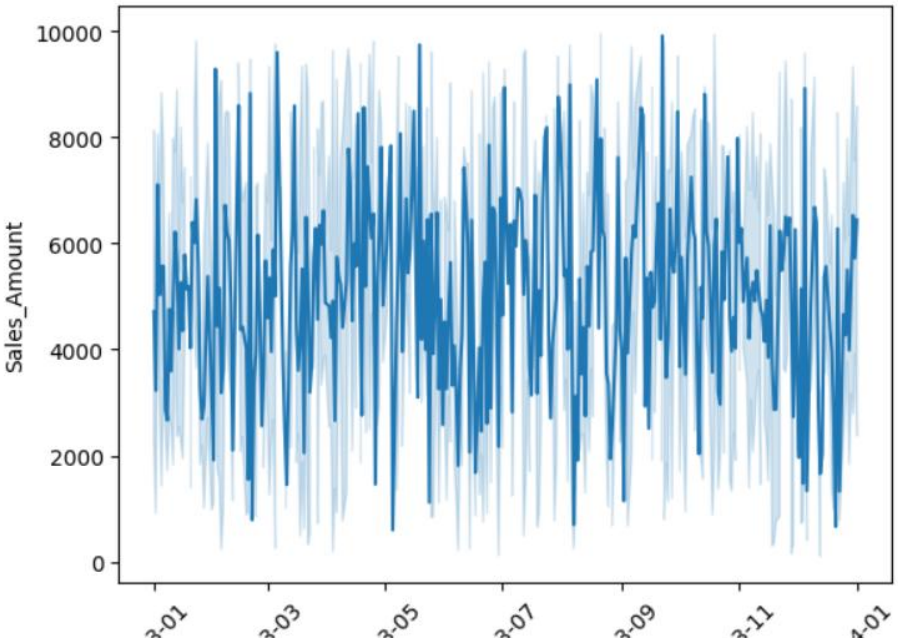
```
Saving sales_data.csv to sales_data (2).csv
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Product_ID           1000 non-null   int64
 1   Sale_Date            1000 non-null   object
 2   Sales_Rep            1000 non-null   object
 3   Region               1000 non-null   object
 4   Sales_Amount         1000 non-null   float64
 5   Quantity_Sold        1000 non-null   int64
 6   Product_Category     1000 non-null   object
 7   Unit_Cost            1000 non-null   float64
 8   Unit_Price           1000 non-null   float64
 9   Customer_Type        1000 non-null   object
 10  Discount             1000 non-null   float64
 11  Payment_Method       1000 non-null   object
 12  Sales_Channel        1000 non-null   object
 13  Region_and_Sales_Rep 1000 non-null   object
dtypes: float64(4), int64(2), object(8)
memory usage: 109.5+ KB
```






1000
0
Furniture
Food
Clothing
Electronics
Product_Category


5000
4000
3000
2000
1000
0
Sales_Amount
North
West
South
East
Region


10000
8000
6000
4000
2000
0
Sales_Amount